

SmallChip AI — 5-min Study Cheat Sheet

Harshith Nelabhotla

Strongsville High School, Strongsville OH 44136, USA

hnelabhotla@students.strongsville.k12.oh.us

Faculty sponsor: Mrs. DiGioia, Strongsville High School Science Research Program

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Read this on the bus ride to school. Memorize these 12 numbers. You'll be asked every one of them.

THE 12 NUMBERS (memorize these)

#	What	Number
1	GCD improvement	99.7% / 370× (3,987,080 → 10,775)
2	GCD timing	0.52 ns WNS, 2097 MHz (identical to OpenROAD)
3	GCD power	1.06 mW (identical to OpenROAD)
4	Held-out test	100% win rate, 87.1% avg, 87.5% median, 66 designs
5	Inference on 15K cells	150ms (CPU)
6	Partial re-place (drag)	14ms for 71 cells
7	Hierarchy proven	30,000,000 cells in 50s
8	Hierarchy projected	100,000,000 cells
9	Model size	18,000 parameters
10	Training data	510 designs (synthetic + ISPD 2005)
11	Per-company value	37,500/yr (1 engineer) to 67,500/yr (5 engineers)
12	License	BSD 3-Clause (commercial use OK, no copyleft)

THE 4 CLAIMS (memorize these)

1. **First free, BSD-3 open-source real-time interactive cell-level chip placement tool** (Chipmind does RTL, Altium/Allegro do PCB, no one does cell-level placement interactively)
2. **First BSD-3 placer with an LLM co-pilot** (no other tool has this)
3. **First end-to-end open-source chip design flow with free interactive placement** (Skywater + Yosys + OpenROAD + KLayout + RISC-V were all open; we fill the placement gap)
4. **First end-to-end proof that interactive placement scales to 30M cells** (50 seconds)

THE 5 NAMES YOU MIGHT BE ASKED ABOUT

- **Cadence Innovus** (500K-1M/yr, batch, 20-60 min)
- **Synopsys IC Compiler II** (500K-1M/yr, batch)
- **OpenROAD** (BSD-3 free, but batch, often diverges on hard designs)
- **DREAMPlace** (academic SOTA, GPU-accelerated, 30 min on V100, batch)
- **RePLAce** (the placer inside OpenROAD; we document it fails to converge on 15K bigblue1)

THE 5 TOOLS YOU MIGHT BE ASKED ABOUT (in adjacent layers)

- **Chipmind** — RTL design with drag-to-replace. Different layer (we do placement, they do RTL).
- **Altium Designer** — PCB design, not chips. Different domain.
- **Cadence Allegro X** — PCB design, not chips.
- **Quadcept** — Schematic capture, not placement.
- **Siemens EDA** — PCB + schematic, not cell-level placement.

THE 3 TOOLS IN OUR ECOSYSTEM (open source)

- **Skywater 130nm** — open PDK
- **Yosys** — open synthesis
- **OpenROAD** — open place-and-route (we complete this layer with interactive)

THE 4 THINGS WE'RE MISSING (honest weaknesses)

1. **5K V3 HPWL plateau at 243K** (V4 retrain fixes this, in progress)
2. **No academic co-author** (50 cold emails planned)
3. **No physical chip in hand** (efabless pending)
4. **No DREAMPlace head-to-head on same hardware** (DREAMPlace not installed locally yet)

THE 5 THINGS YOU MUST SAY IF ASKED "WHAT IS NEW?"

1. "First free real-time interactive cell-level placer" (no commercial or academic tool has this for placement)
2. "First BSD-3 placer with an LLM co-pilot"
3. "First end-to-end open-source chip design flow with free interactive placement"
4. "First interactive placement proven to scale to 30 million cells"
5. "99.7% GCD improvement, identical timing + power, validated by OpenROAD's own analyzer"

THE PITCH (60 sec, memorize)

"Every chip in the world needs placement. Industry tools cost 500K to 1M per year per seat. The open-source tool, OpenROAD, is free but it runs as a batch job that takes 5 to 30 minutes per placement and often diverges on hard designs. There is no free, real-time, interactive placer for cells on a chip die.

I'm a 14-year-old and I built SmallChip AI. It uses a graph attention network with 18,000 parameters — small enough to run on a laptop, fast enough to give a 150-millisecond response. You can drag a cell on a canvas and see the chip re-place in 14 milliseconds. You can type 'make this faster' in plain English and the chip re-places.

On the GCD benchmark with 734 cells, after OpenROAD's own legalization and static timing analysis, our placement achieves 99.7% wirelength reduction — that's 370 times better — with identical timing and identical power. On a clean held-out test of 66 designs, we win on 100% of them, with 87.1% average improvement.

The architecture scales to 30 million cells, proven end-to-end. The whole thing is BSD-3 open source on GitHub, free for anyone to use. We're the missing piece in the open-source EDA ecosystem — Skywater, Yosys, OpenROAD, KLayout, and RISC-V were all open, but no one had a free interactive placer until now."

THE Q&A SHORTCUTS

"How is this different from Chipmind?" "Chipmind does RTL design — they help you write Verilog by dragging modules. We do cell-level placement — we put cells on the die after the code is written. Different layer, different problem. Both are AI for chip design, but they're complementary."

"How is this different from OpenROAD?" "OpenROAD is a full RTL-to-GDS flow with batch placement. We do real-time interactive placement. We're complementary — for detailed routing and CTS, OpenROAD is still the way. For interactive placement, we win."

"How is this different from DREAMPlace?" "DREAMPlace is GPU-accelerated batch placement. ~30 minutes on a V100 for 211K-cell adaptec1. We're CPU + small model, real-time, with an LLM co-pilot. Different optimization regime."

"How do you know your test isn't contaminated?" "We use a hash-based 80/20 split — each design gets a SHA-256 hash, we sort by hash, take the bottom 20% as held-out. We verified 0% overlap with training before reporting the 100% number. We caught and removed an earlier 75.2% number on 91 designs that turned out to be 100% contaminated. The current 87.1% is on a clean test."

"What is your biggest weakness?" "5K V3 HPWL plateau at 243K with cell_w=6 microns. V4 multi-objective retrain in progress. Also: no academic co-author yet, no physical chip in hand, no DREAMPlace head-to-head on same hardware. All of these are being worked on."

NUMBERS YOU MIGHT BE ASKED

- 8,000× faster than industry: based on 30 min industry vs 150ms ours on 15K cells
- 5K-15K scaling: 355K-540K HPWL, 41-85 $\mu\text{m}/\text{net}$ (monotonically decreasing)
- 66 designs in held-out: 8 small (<200 cells, 93.9% avg), 56 medium (200-600, 86.1% avg), 2 large (≥ 600 , 88.0% avg)
- 30M scaling: 1,281,714 DBU/net, 50.6s workers, 12GB peak memory
- BSD-3 license: no copyleft, no "share-alike", use commercially for free
- Per-company value: 37,500/yr (1 engineer) to 67,500/yr (5 engineers), \$200K/yr (10-student university lab)

WHAT TO DO IF YOU DON'T KNOW

"It's a good question. I haven't measured that specifically, but the closest data we have is [X]. I'd want to come back with a real number rather than guess."

NEVER make up a number. NEVER say "I think" without a citation. If you don't know, say so and offer to follow up.

AFTER ISEF (if you win, what comes next)

- V4 retrain
- 100M head-to-head with DREAMPlace
- efabless chip tapeout
- 1+ academic co-author
- 5+ beta testers
- arXiv submission
- Demo video
- 50 cold emails
- ISEF paper polish
- NEOSEF Feb 2027 → ISEF May 2027